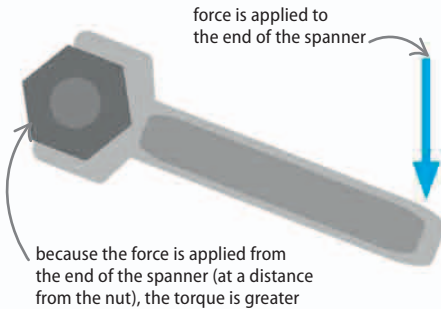


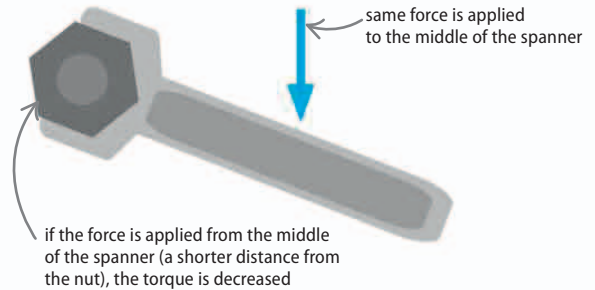
Torque

This term refers to the turning effect of a force—its ability to create rotation rather than linear (straight-line) motion. Torque is dependent on the size of a force and its distance from the turning point, or pivot. Forces applied farther from the pivot result in a larger torque. The torque (or moment) of a force is calculated by multiplying force and distance.



△ Large torque

Applying a force to the end of a spanner handle maximizes its torque, making it easier to undo a stiff nut.



△ Small torque

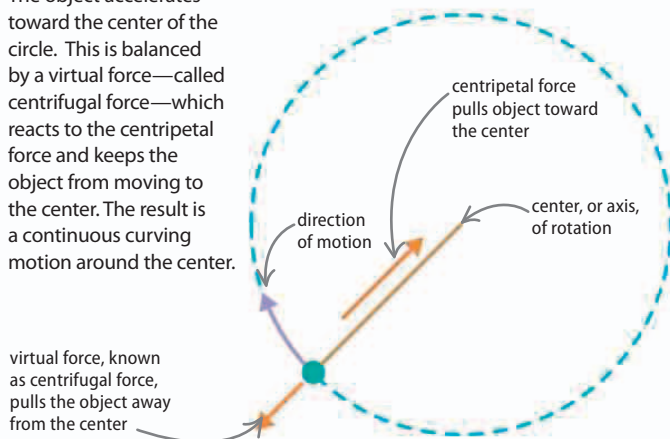
Applying the same force to halfway along the handle results in half the torque, so turning the nut requires more effort.

Rotational motion

When an object moves in a circle, it is acted on by two forces. The centripetal force pulls it toward the center, such as gravity on a space satellite or the strength of a string attached to a ball. A second centrifugal force counteracts the centripetal force by pulling the object away from the center.

▷ In a spin

The object accelerates toward the center of the circle. This is balanced by a virtual force—called centrifugal force—which reacts to the centripetal force and keeps the object from moving to the center. The result is a continuous curving motion around the center.



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Angular momentum

Any spinning object has what is called angular momentum, which is proportional to the mass of the object, its rotational speed, and the average distance of the mass from the center of the spin. The ice skater uses this phenomenon to control the speed of her spins. When she stretches out her arms, she spreads her mass over a wider area, which creates a relatively slow rate of spin. When she draws them back in, all her mass is centered over the axis of rotation, and she spins faster.

